

G27 Pull Request Reliability Review

5 October 2026 | Findings and verified integration changes

We reproduced the combined-branch failure and repaired the source receipt by rerunning the frozen analysis. Normal prediction parity had missed error paths; passing branches separately had missed their integration. The numerical findings remain unchanged.

Finding	Cause and response
PR5 model state	State advanced before the classifier returned. Included PR8 commits state only after success and restores it after a failed batch.
PR8 EEG stream	Model rollback did not restore consumed EEG. Updated PR8 fails closed: a processing error makes the decoder terminal. Recovery reconstructs it and replays from a known run boundary; chunk retry is unsupported.
PR7/8 source receipts	PR7 rejected the changed Riemannian source; PR8's test ignored extra stream-source hashes. Actual reruns preserve historical evidence. Tests now verify all declared current sources, output hashes and prediction correctness.
PR6 evidence labels	Random balanced subsets were easy to mistake for chronological calibration. PNG, SVG and HTML captions now state the limitation. Historical receipts and convenience exports have an explicit scope.

Verification and prevention

127 passed, 1 skipped in 100.31s (0:01:40). Preflight and published-result arithmetic pass. All 24 early choices, 1,920 outer and 1,080 inner predictions match; 3,840 historical forward predictions and 192 summaries reproduce. Automated checks run on pull requests, pushes and merge groups without raw EEG. GitHub: 130 passed in 35.78s on d1a84f2 (Python 3.11, optional accuracy dependencies installed).

Scientific result and limits

At 60 calibration labels and +3.5 seconds, fixed CSP scores **82/120** versus **75/120** for selection; P2 PRE falls from 13/20 to 8/20. These exposed retrospective tails are not independent validation. Feedback onset and clinical side remain uncertain. Repairing reproducibility does not establish motor-intent specificity or rehabilitation benefit.

Sources: Anna's comments on PR5, PR8, PR7 and PR6. Requested independent reviewer: Astra / Ultra; runtime identity unverified. Exact hashes: results/pr_reliability/verification.json. Candidate integration only; no main merge or competition submission.